Linux OS and Scripting – Class 1

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Part A — User and System Investigation

Determine the username of the account you are currently using.

Display your UID, primary GID, and supplementary groups.

Display all users currently logged in to the system.

Determine what activities the currently logged-in users are performing.

Examine /etc/passwd and identify the following information for your current account:

Username

UID

GID

Home directory

Login shell

Determine how many user-account entries are present in /etc/passwd.

Display only the username and UID fields from /etc/passwd.

Find all accounts whose UID is 0. Explain the security significance of UID 0.

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```
┌──(aduuu㉿kali)-[~]
└─$ whoami
aduuu

┌──(aduuu㉿kali)-[~]
└─$ id
uid=1000(aduuu) gid=1000(aduuu) groups=1000(aduuu),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plu
gdev),100(users),101(netdev),102(scanner),968(vboxsf),979(bluetooth),999(lpadmin)

┌──(aduuu㉿kali)-[~]
└─$ who
aduuu    seat0        2026-09-01 20:40 (:0)

┌──(aduuu㉿kali)-[~]
└─$ w
 20:41:34 up 1 min,  1 user,  load average: 0.77, 0.34, 0.12
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
aduuu             -                20:40           0.00s  0.01s lightdm --session-child 13 24
```

```
┌──(aduuu㉿kali)-[~]
└─$ grep "^$(whoami):" /etc/passwd
aduuu:x:1000:1000:aditi,,,:/home/aduuu:/bin/bash

┌──(aduuu㉿kali)-[~]
└─$ wc -l /etc/passwd
60 /etc/passwd

┌──(aduuu㉿kali)-[~]
└─$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
```

```
┌──(aduuu㉿kali)-[~]
└─$ awk -F: '$3 == 0 {print $1}' /etc/passwd
root

┌──(aduuu㉿kali)-[~]
└─$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
```

```
┌──(aduuu㉿kali)-[~]
└─$ sudo cat /etc/shadow
[sudo] password for aduuu:
root:!:20674:0:99999:7:::
daemon:*:20674:0:99999:7:::
bin:*:20674:0:99999:7:::
```

/etc/passwd contains basic user information such as username, UID, GID, home directory and login shell. /etc/shadow stores password hashes and other password-related information and is accessible only to privileged users. Password hashes are stored in /etc/shadow to protect them from being accessed by ordinary users.

Part B — Creating and Verifying Users

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

Verify that all three accounts have been added to /etc/passwd.

Determine the UID assigned to each newly created user.

Determine the home directory and default shell assigned to alice.

Inspect /home and verify that the required home directories were created.

Set passwords for all three accounts.

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
┌──(aduuu㉿kali)-[~]
└─$ ls /home
aduuu  alice  bob  charlie

┌──(aduuu㉿kali)-[~]
└─$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh
bob:x:1002:1002::/home/bob:/bin/sh
charlie:x:1003:1003::/home/charlie:/bin/sh

┌──(aduuu㉿kali)-[~]
└─$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice)

┌──(aduuu㉿kali)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)

┌──(aduuu㉿kali)-[~]
└─$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie)

┌──(aduuu㉿kali)-[~]
└─$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh
bob:x:1002:1002::/home/bob:/bin/sh
charlie:x:1003:1003::/home/charlie:/bin/sh

┌──(aduuu㉿kali)-[~]
└─$ grep "^alice:" /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh

┌──(aduuu㉿kali)-[~]
└─$ ls /home
aduuu  alice  bob  charlie

┌──(aduuu㉿kali)-[~]
└─$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

┌──(aduuu㉿kali)-[~]
└─$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully
```

```
┌──(aduuu㉿kali)-[~]
└─$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully

┌──(aduuu㉿kali)-[~]
└─$ su - alice
Password:
$ whoami
alice
$ exit
```

Part C — Group Management and Access Control

Create two groups:

security and developers

Add alice and charlie to the security group.

Add bob to the developers group.

Without switching accounts, verify the group membership of all three users.

Display the entries corresponding to security and developers from /etc/group.

Add bob to the security group without removing him from the developers group. Verify that bob now belongs to both groups.

```
┌──(aduuu㉿kali)-[~]
└─$ sudo groupadd security

┌──(aduuu㉿kali)-[~]
└─$ sudo groupadd developers

┌──(aduuu㉿kali)-[~]
└─$ sudo usermod -aG security alice

┌──(aduuu㉿kali)-[~]
└─$ sudo usermod -aG security charlie

┌──(aduuu㉿kali)-[~]
└─$ sudo usermod -aG developers bob

┌──(aduuu㉿kali)-[~]
└─$ groups alice
alice : alice security

┌──(aduuu㉿kali)-[~]
└─$ groups bob
bob : bob developers

┌──(aduuu㉿kali)-[~]
└─$ groups charlie
charlie : charlie security

┌──(aduuu㉿kali)-[~]
└─$ grep -E '^(security|developers):' /etc/group
security:x:1004:alice,charlie
developers:x:1005:bob

┌──(aduuu㉿kali)-[~]
└─$ sudo usermod -aG security bob

┌──(aduuu㉿kali)-[~]
└─$ groups bob
bob : bob security developers
```